

SEMSAI

Your Portfolio

68

Portfolio Health

Moderate

↑ 6 from last decision

Cashflow 72/100



Risk 58/100



Liquidity 65/100



Flexibility 78/100



 How is this calculated?

Phase 1:

Input Structure and Calculation Framework

1. User Inputs (Data Payload)

1.1 Investor Financial Profile

Income

- Net Monthly Income
- Income Type (Salary / Freelance)

Obligations

- Monthly Fixed Expenses

Liquidity

- Available Cash
- Credit Access (Yes / No)

Risk Appetite

- Risk Profile (Conservative / Moderate / Aggressive)
 - Maximum Acceptable Installment Ratio (e.g., 40% = 0.40)
-

1.2 Portfolio Data (Per Unit Owned)

For each property unit:

- Monthly Installment
- Remaining Balance
- Current Market Value

From all units, calculate:

Total Installments = Sum of all Monthly Installments

2. Intermediate Financial Calculations

These variables are calculated internally by the system.

2.1 Free Cashflow

Free Cashflow = Net Monthly Income – Monthly Fixed Expenses – Total Installments

This represents how much money remains at the end of each month.

2.2 Debt-to-Income Ratio (DTI)

DTI = Total Installments ÷ Net Monthly Income

This measures how much of the investor's income is allocated to property payments.

2.3 Liquidity Runway (Months)

$\text{Runway (Months)} = \text{Available Cash} \div (\text{Monthly Fixed Expenses} + \text{Total Installments})$

This measures how many months the investor can survive if income stops completely.

2.4 Risk Buffer

$\text{Risk Buffer} = \text{Maximum Acceptable Installment Ratio} - \text{DTI}$

If the value is:

- Positive → Investor is within safe leverage range
 - Zero → Investor is at their maximum risk tolerance
 - Negative → Investor is overleveraged
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3. Score Normalization (0–100 Scale)

Each financial dimension is converted into a score between 0 and 100.

3.1 Cashflow Score (30% Weight)

Target: Keeping 40% of income free is considered ideal.

$\text{Cashflow Score} = \text{Max}(0, \text{Min}(100, (\text{Free Cashflow} \div (\text{Net Income} \times 0.40)) \times 100))$

If Free Cashflow is negative, the score is automatically 0.

3.2 Risk Score (30% Weight)

$\text{Risk Score} = \text{Max}(0, \text{Min}(100, (\text{Risk Buffer} \div \text{Maximum Acceptable Installment Ratio}) \times 100))$

Income Stability Adjustment:

If Income Type = Freelance → subtract 15 points from Risk Score.

After adjustment, clamp the score between 0 and 100.

3.3 Liquidity Score (20% Weight)

Assumption: 6 months of runway = perfect score.

Liquidity Score =
 $\text{Min}(100, (\text{Runway} \div 6) \times 100)$

If Runway $\geq$ 6 months $\rightarrow$ Liquidity Score = 100.

3.4 Flexibility Score (20% Weight)

Heuristic model:

Start with baseline score = 50.

Add:

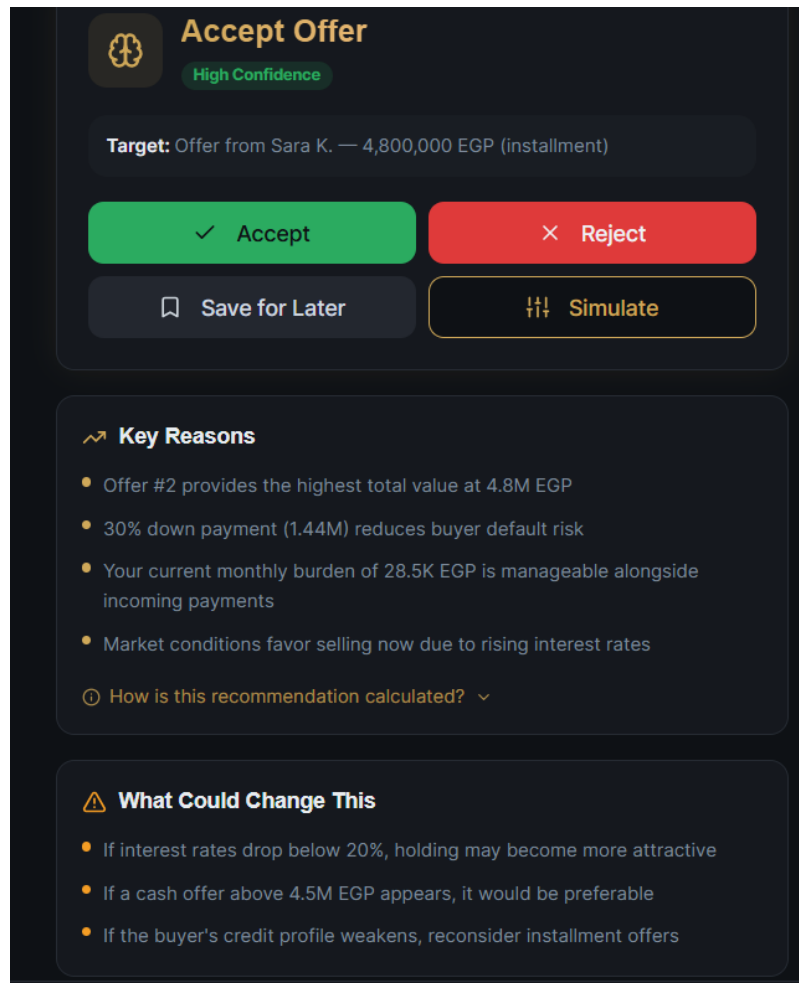
- +10 points per owned unit (maximum +30)
- +25 points if Credit Access = Yes

Flexibility Score = Minimum of 100 and total calculated value.

4. Final Portfolio Health Score

Portfolio Score =
 $(0.30 \times \text{Cashflow Score})$

- $(0.30 \times \text{Risk Score})$
- $(0.20 \times \text{Liquidity Score})$
- $(0.20 \times \text{Flexibility Score})$



Overview

The system analyzes all available offers using a multi-factor decision model that evaluates financial value, risk exposure, seller liquidity position, and market conditions. Based on these dimensions, it generates a final recommendation (Accept, Reject, or Hold) along with a confidence score.

1. Financial Value Assessment

The system first compares offers based on **total monetary value**.

For installment offers, the total contract value is calculated as:

Total Value = Down Payment + Sum of Installments

The offer with the highest financial return receives a higher base score.

However, installment offers are not evaluated at face value only — future payments are discounted using time value of money principles when applicable.

2. Risk Evaluation (Buyer Default Risk)

The system evaluates the likelihood of payment default based on:

- Down payment percentage
- Installment duration
- Buyer financial profile (if available)

Higher down payments reduce default probability and increase the offer's adjusted score.

Example logic:

- 30%+ down payment → Low risk +10
- 15–30% → Medium risk +5
- Below 15% → Higher risk +0

$$\text{Risk Adjustment Multiplier} = 1 + 100 / \text{Risk Points}$$

Risk directly affects the final offer ranking.

3. Seller liquidity factor

Financial Pressure Multiplier

High	0.9
Medium	1.0
Low	1.1

$$\text{PV after Seller Adjustment} = \text{PV of Offer} \times \text{Seller Liquidity Factor}$$

4. Market Condition Analysis

Macroeconomic indicators are incorporated into the decision:

- Interest rate trends

- Inflation levels
- Real estate liquidity trends

For example:

- Rising interest rates → Future demand may decrease → Favor selling now +10
- Falling interest rates → Market may improve → Holding could be beneficial +0

This aligns the recommendation with real market timing.

5. Confidence Score Calculation

The system assigns a confidence level (Low, Medium, High) based on:

Final Score=(PV of Offer)×(Risk Adjustment Multiplier)×(Seller Liquidity Factor)+(Market Factor)

